

PHYS 241: Signal Processing

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Abstract

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Contents

1	Introduction	1
2	Foundations	1
2.1	Waves	1
2.1.1	Properties of Waves	1
2.1.2	Waveforms	1
2.2	Linear Systems	2
3	Electromagnetic Theory	2
3.1	Electric Charge	3
3.2	Electric Field	3
3.3	Electric Potential	3
3.4	Electric Potential Energy	4
3.5	Magnetic Field	4
3.6	Magnetic Flux	5
3.7	Electromagnetic Induction	5
3.7.1	Faraday's Law	5
3.7.2	Lenz's Law	5
3.7.3	Mutual Inductance	6
4	Circuit Fundamentals	6
4.1	Current and Voltage	6
4.2	Electromotive Force	7
4.3	Ohm's Law	7
4.4	Kirchhoff's Laws	8
4.5	Direct and Alternating Current	9
4.6	Static Circuits	10
4.7	Circuit Classification	10
5	Circuit Components	11
5.1	Resistors	11
5.1.1	Temperature Dependence	11
5.1.2	Power Dissipation	12
5.2	Batteries	12
5.3	Ground and Reference	12
5.4	Capacitors	13
5.4.1	Dielectric Insulator	13
5.4.2	Energy Stored in a Capacitor	13
5.4.3	Charging a Capacitor	14
5.5	Inductors	14
5.5.1	Energy Storage	15
5.5.2	Networks of Inductors	16
5.5.3	Solenoid	16
5.6	Transformers & Mutual Inductance (Not Tested)	16

6	Circuit Analysis	18
6.1	RC Circuit	18
6.1.1	DC Analysis	18
6.1.2	AC Analysis	19
6.2	LR Circuit	20
6.2.1	DC Analysis	21
6.2.2	AC Analysis	22
6.3	Frequency Behaviour Summary	23
7	Phasors and Complex Impedance	23
7.1	Complex Representation of AC Signals	23
7.2	Working with Complex Signals	24
7.3	Capacitor Current	24
7.4	Impedance	24
7.5	Power Dissipation of AC Signals	26
7.6	Sinusoidally Driven RC Circuit	26
7.7	Transfer Function	27
7.8	Filters and Bode Plots	27
8	RLC Circuits	30
8.1	Charging with a Step Voltage	30
8.2	Solution by Characteristic Equation	31
8.3	Damping Cases	31
8.3.1	Underdamped ($\Gamma < 2\omega_0$).	31
8.3.2	Overdamped ($\Gamma > 2\omega_0$).	32
8.3.3	Critically damped ($\Gamma = 2\omega_0$).	32
8.4	RLC Circuit with AC Driving	32
8.5	Quality Factor	33
8.6	Power and Resonance	33
8.7	Capacitor Voltage in a Driven RLC Circuit	34
8.7.1	Limiting Behaviour	34
8.8	Quality Factor	35
8.8.1	Derivation for a Series RLC Circuit	35
8.9	Quality Factor and Energy Decay	35
8.10	Power Input into a Driven RLC Series Circuit	36
8.10.1	Steady State	36
8.10.2	Bandwidth and Quality Factor	37
9	Circuit Network Analysis	37
9.1	Kirchhoff's Laws (Review)	37
9.2	Terminology	37
9.3	Methods of Analysis	37
9.3.1	Branch Method	37
9.3.2	Loop Current Method	38
9.3.3	Nodal Method	38
9.4	Superposition Theorem	38

9.5	Current Sources	38
9.6	Converting Between Source Types	39
9.7	Thévenin's Theorem	39
9.7.1	Determining e and r	39
9.8	Norton's Theorem	40
9.8.1	Equivalence of Thévenin and Norton Representations	40
10	Fourier Analysis	40
10.1	Fourier Series	40
10.2	Orthogonality Relations	41
10.3	Fourier Coefficients	41
10.4	Convergence: Dirichlet's Theorem	42
10.5	Fourier Transform	42
10.6	Gibbs Phenomenon	42
10.7	Example: Sawtooth Wave	43
10.8	Even and Odd Functions	43
10.9	Transfer Functions and Linear Systems	43
10.9.1	Application to a Sawtooth Input	44
10.10	Complex Form of the Fourier Series	44
10.11	Fourier Series vs. Fourier Transform	45
10.12	From Fourier Series to Fourier Transform	45
10.12.1	Negative Frequencies	45
10.12.2	The Limiting Argument	45
10.13	Fourier Transform Pairs	46
10.13.1	Example: Fourier Transform of a Box Function	46
10.14	Time-Bandwidth Product	46
10.15	Dirac Delta Function	47
10.16	Application: Pulse Input into an RC Circuit	47
10.17	Translated Dirac Delta Function	47
10.18	Fourier Transforms of Cosine and Sine	48
10.19	Fourier Transform Pairs	48
10.20	Translation Theorems	49
10.21	Example: Fourier Transform of a Pulse Pair	49
10.22	Reality Condition	49
10.23	Power Spectrum	49
10.23.1	One-Sided Spectrum	50
10.24	Finite-Length Signals and Windowing	50
10.25	Discrete Sampling and the Nyquist Criterion	50
10.26	Discrete Fourier Transform	51
10.27	Amplitude Modulation	51
11	Appendix	51
12	Useful Links	51

1 Introduction

2 Foundations

2.1 Waves

2.1.1 Properties of Waves

To describe waves, considering a sinusoidal wave of the form $A_p \sin(2\pi\nu t)$, we use the following terms:

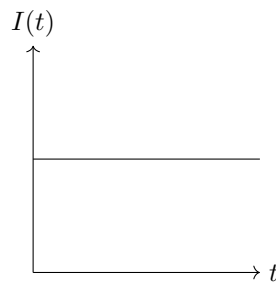
- Peak amplitude (A_p): maximum value of the wave from its equilibrium position.
- Peak-to-peak amplitude: total height of the wave from its maximum to its minimum value (i.e., $2A_p$).
- Frequency (ν): number of cycles per second (Hz).
- Time (t): time variable.

To describe AC signals effectively, we take the root mean square (RMS) amplitude $\frac{A_p}{\sqrt{2}}$ of values such as current and voltage. This is because the mean value of an AC signal is zero, and the RMS is a more meaningful measure of the signal's amplitude as it gives a positive value that is proportional to the energy in the signal.

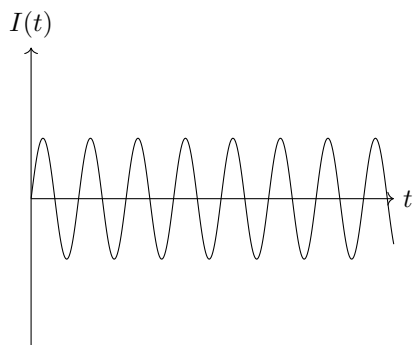
2.1.2 Waveforms

Definition 1 (Waveform). *A waveform is a graphical representation of how a signal varies over time, and can take various forms such as a sine wave, square wave, triangle wave, or sawtooth wave.*

1. Direct current (DC). A direct current is constant in time, so its graph is a horizontal line.



2. Alternating current (AC). An alternating current varies periodically in time and typically oscillates about zero in a sinusoidal pattern.



3. Pulsating current (DC + AC). A pulsating current is an alternating current superimposed on a non-zero DC level. This is just a matter of adding a constant DC level to an AC signal so that the signal is either shifted up or down with respect to the zero current level.

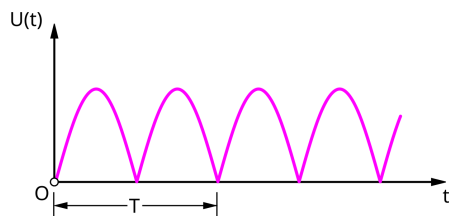


Figure 1: Pulsating direct current diagram.

2.2 Linear Systems

Definition 2 (Linear System). *A system H is linear if it satisfies superposition*

$$H[x_1(t) + x_2(t)] = H[x_1(t)] + H[x_2(t)],$$

and scaling

$$H[\alpha x(t)] = \alpha H[x(t)],$$

*for all inputs $x_1(t), x_2(t)$ and constant α .*¹

3 Electromagnetic Theory

Honestly, this section there isn't really something that he will test you upon but it should be knowledge that you should know to better grasp the idea behind how a circuit works.

¹PS: this definition is analogic to the definition of a linear transform from MATH 223

3.1 Electric Charge

Definition 3 (Electric Charge). *Charge is a fundamental physical property that comes in two types: positive (+) and negative (-) (which cancel). Opposite charges attract and equal amounts of positive and negative charge cancel, so most matter is electrically neutral.*

Proposition 1 (Conservation of Charge). *Electric charge is conserved, meaning it cannot be created or destroyed, only transferred.*

Proposition 2 (Quantization of Charge). *Electric charge is also quantized, existing in discrete multiples of the elementary charge*

$$e = 1.6 \times 10^{-19} \text{ C},$$

where the coulomb (C) is the SI unit of charge.

3.2 Electric Field

Definition 4 (Electric Field). *The electric field $\vec{E}$ is a vector field produced by electric charges. It describes the force that a charge would experience at any point in space. Quantitatively, the electric field at a point is defined as the force $\vec{F}$ acting on a small positive test charge q placed at that point, divided by the magnitude of the charge:*

$$\vec{E} = \frac{\vec{F}}{q}.$$

Proposition 3 (Direction of Electric Field). *The direction of $\vec{E}$ is the direction of the force on a positive test charge.*

Remark 1. *The SI units for an electric field are newtons per coulomb (N/C), which are equivalent to volts per meter (V/m).*

3.3 Electric Potential

Definition 5 (Electric Potential). *The electric potential is a scalar quantity with units of volts (V) that represents the work done per unit charge to move a positive test charge from a reference point (often taken at infinity) to the point $\vec{r}$, without changing its kinetic energy.*

Proposition 4 (Formula for Electric Potential). *The electric potential $V(\vec{r})$ at a point is the electric potential energy per unit charge:*

$$V = \frac{U}{q}.$$

Proposition 5 (Relation to Electric Field). *It is also defined relative to a reference point $\vec{r}_0$ by*

$$V(\vec{r}) - V(\vec{r}_0) = - \int_{\vec{r}_0}^{\vec{r}} \vec{E} \cdot d\vec{l},$$

where $\vec{E}$ is the electric field. The electric field is related to the potential by

$$\vec{E} = -\nabla V.$$

Example (Uniform Electric Field). If $\vec{E}$ is uniform, the potential difference between two points separated by $\vec{d}$ is

$$\Delta V = -\vec{E} \cdot \vec{d}.$$

3.4 Electric Potential Energy

Definition 6 (Electric Potential Energy). *Electric potential energy U is the energy a charge possesses due to its position in an electric field. Since electric potential is defined relative to a reference point, the potential energy is also defined relative to that same reference.*

Proposition 6 (Formula for Electric Potential Energy). *For a charge q at a point where the electric potential is V , the potential energy is*

$$U = qV,$$

and is measured in joules (J).

An electron placed between two charged plates experiences a potential difference ΔV . The change in its electric potential energy is

$$\Delta U = q \Delta V.$$

If the electron is allowed to move freely, this change in electric potential energy is converted into kinetic energy:

$$\Delta K = -\Delta U.$$

3.5 Magnetic Field

Definition 7 (Magnetic Field). *A magnetic field $\vec{B}$ is a vector field produced by moving electric charges (currents) and intrinsic magnetic moments of particles. It exerts a force on moving charges given by the Lorentz force law, which is perpendicular to both the velocity and the magnetic field.*

Remark 2 (Magnetic Fields vs. Electric Fields). *Unlike electric fields, which act on charges whether stationary or moving, magnetic fields act only on moving charges.*

Theorem 1 (Lorentz Force Law). *The force $\vec{F}$ on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given by*

$$\vec{F} = q(\vec{v} \times \vec{B}).$$

Proposition 7 (Magnetic Field from a Current-Carrying Wire). *For a long straight current-carrying wire, the magnitude of the magnetic field at a distance r is*

$$B = \frac{\mu_0 I}{2\pi r},$$

where μ_0 is the permeability of free space² and I is the current. The SI unit of magnetic field is the tesla (T).

3.6 Magnetic Flux

Definition 8 (Magnetic Flux). *Magnetic flux Φ_B through a surface is defined as the surface integral of the magnetic field $\vec{B}$ over that surface:*

$$\Phi_B = \int \vec{B} \cdot d\vec{A},$$

where $d\vec{A}$ is an infinitesimal area element of the surface, and the dot product accounts for the angle between the magnetic field and the normal to the surface.

Remark 3. *The SI unit of magnetic flux is the weber (Wb).*

3.7 Electromagnetic Induction

3.7.1 Faraday's Law

Theorem 2 (Faraday's Law). *Faraday's Law states that a changing magnetic flux through a closed loop induces an electromotive force (EMF) in the loop:*

$$\mathcal{E}_{EMF} = -\frac{d\Phi_B}{dt},$$

where $\mathcal{E}_{EMF}$ is the induced EMF and Φ_B is the magnetic flux through a wire loop. The negative sign indicates that the induced EMF creates a current whose magnetic field opposes the change in flux, as described by Lenz's Law.

3.7.2 Lenz's Law

Law 1 (Lenz's Law). *The induced current in a circuit flows in a direction such that the magnetic field it produces opposes the change in magnetic flux that generated it.*

Consider any closed conducting loop. If the magnetic flux through the loop increases, the induced current produces a magnetic field that opposes the increase. If the flux decreases, the induced current produces a field that tends to restore it.

When the magnetic flux is increasing, the induced current is directed opposite to the orientation predicted by the right-hand rule for the increasing field.

²The permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$.

When the flux is decreasing, the induced current follows the orientation that attempts to reinforce the original field.



Figure 2: Direction of the induced current according to Lenz's Law.

3.7.3 Mutual Inductance

Definition 9 (Mutual Inductance). *Mutual inductance is the property of two nearby circuits whereby a time-varying current in one coil induces an electromotive force (EMF) in another through magnetic flux linkage.*

If a current $I_1(t)$ flows in the primary coil, it produces a magnetic flux Φ_{21} through the secondary coil. According to Faraday's law,

$$\mathcal{E}_2 = -\frac{d\Phi_{21}}{dt}.$$

The mutual inductance M is defined by

$$\Phi_{21} = MI_1,$$

which gives

$$\mathcal{E}_2 = -M\frac{dI_1}{dt}.$$

The value of M depends on the number of turns in each coil, their geometry, their separation, and the magnetic permeability of the material between them.

4 Circuit Fundamentals

4.1 Current and Voltage

Definition 10 (Electric Current). *Electric current is the flow of electrons through a wire driven by an electric field potential energy difference.*

Proposition 8. *The current is defined as the rate of charge past a point in a circuit:*

$$I = \frac{dQ}{dt}, \tag{1}$$

and is measured in amperes (A), where $1 A = 1 C/s$.

Definition 11 (Voltage). *The voltage (or electric potential difference) is the difference in potential energy between two points in an electric field.*

Definition 12 (Electron-volt). *An electron-volt (eV) is a unit of energy equal to the energy gained or lost by an electron when it moves through an electric potential difference of one volt.*

Proposition 9. *Since energy is given by $U = qV$, for a charge equal to the elementary charge e , we have*

$$1 \text{ eV} = e \cdot 1 \text{ V} = 1.602 \times 10^{-19} \text{ J}.$$

When analyzing circuits, current is defined using the convention of positive charge flow. By convention, current flows from higher potential to lower potential, even though in metallic conductors the actual charge carriers are electrons moving in the opposite direction.

When the circuit is closed, electrons flow externally from the anode to the cathode, while conventional current flows from the cathode to the anode. As the charges move through the circuit, the battery's chemical energy is converted into electrical energy and then into other forms such as heat or light.

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2026

4.2 Electromotive Force

Definition 13 (Electromotive Force (EMF)). *Electromotive Force (EMF)³ is the energy supplied per unit charge by an energy source such as a battery or generator. It represents the work done by the source to move a charge around a complete circuit. EMF is measured in volts (V).*

To better understand the concept of EMF, consider the following analogy: EMF is like a voltage credit that is then all spent by dropping it over the elements of the circuit. Think of your checking account with two columns: credit (money in) and debit (money spent). EMF is the “credit” that is then all “spent” by dropping it over the elements of the circuit.

4.3 Ohm's Law

Definition 14 (Ohm's Law). *Ohm's Law states that the current I flowing through a conductor is directly proportional to the voltage V across it and inversely proportional to its resistance R :*

$$I = \frac{V}{R}.$$

Microscopically, Ohm's Law can be understood in terms of the motion of electrons in a conductor. When a voltage is applied across a conductor, it creates an electric field that exerts a force on the free electrons, causing them to

³EMF is the total energy supplied per unit charge by a source. The terminal voltage across a real source is typically less than the EMF due to internal resistance.

move and create an electric current. The resistance of the conductor arises from collisions between the electrons and the atoms in the material, which impede their motion.

The current density J is related to the electric field strength E by

$$J = \sigma E, \quad E = J \rho_r,$$

where σ is the conductivity and ρ_r is the resistivity. The current is

$$I = AJ = A\sigma E.$$

Using $E = \Delta V/l$,

$$I = \frac{A\sigma \Delta V}{l} = \frac{A \Delta V}{\rho_r l}.$$

From Ohm's Law, $\Delta V = IR$, which gives the resistance in terms of material properties:

$$R = \frac{\rho_r l}{A}.$$

Ohm's Law can be represented graphically using the Ohm's Law Wheel, which illustrates the relationships between current, voltage, resistance, and conductance.

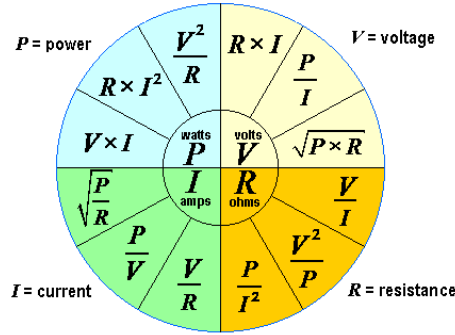


Figure 3: Ohm's Law Wheel and Current Density.

4.4 Kirchhoff's Laws

Definition 15 (Kirchhoff's Voltage Law (KVL)). *The algebraic sum of all potential differences around any closed loop in a circuit is equal to zero.*

KVL allows us to determine the potential difference between two points in a circuit by accounting for the voltage changes across each component along the path.

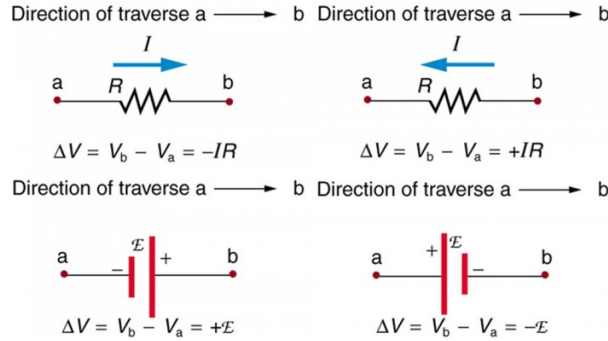


Figure 4: Each resistor and voltage source is traversed from a to b . The corresponding potential changes across each element are indicated. (The script $\mathcal{E}$ denotes the emf.)

Another common circuit element is the inductor. In the context of KVL, the voltage across an inductor is included in the loop equation in the same way as the voltage across resistors or capacitors.

Definition 16 (Kirchhoff's Current Law (KCL)). *The sum of the currents entering a node is equal to the sum of the currents leaving the node.*

4.5 Direct and Alternating Current

Definition 17 (Direct Current (DC)). *Direct Current (DC) is a form of current where voltage and current are constant over time; commonly found in batteries.*

Definition 18 (DC Offset). *A DC offset is the addition of a constant DC value to an AC signal. This shifts the entire signal up or down relative to the 0 V level, without changing the shape of the AC signal.*

Definition 19 (Alternating Current (AC)). *Alternating Current (AC) is a form of current that changes over time, often in a sinusoidal manner. AC currents are commonly used in power distribution systems, such as household electrical outlets.*

In general, an alternating current source is represented in circuit diagrams by a circular source symbol containing a sinusoidal waveform.

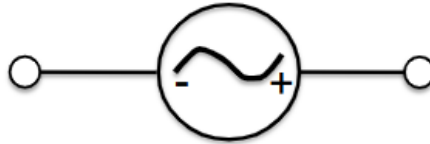


Figure 5: Alternating Current symbol.

If the current is sinusoidal, it is commonly written as

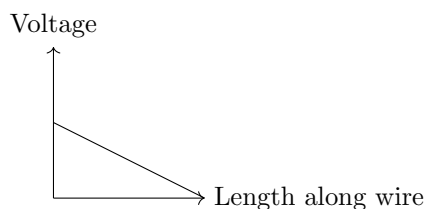
$$I(t) = I_0 \sin(\omega t),$$

where I_0 is the amplitude and ω is the angular frequency.

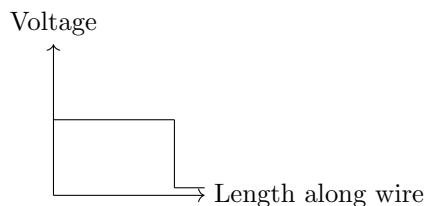
4.6 Static Circuits

Consider the variation of electric potential along a conducting wire.

If the wire has a small but non-negligible resistance that is uniformly distributed, the electric potential decreases approximately linearly with distance, since the voltage drop across a uniform conductor is proportional to its length.



In contrast, if a discrete resistor with resistance much larger than that of the wire is introduced, nearly all of the voltage drop occurs across the resistor. The potential along the wire remains approximately constant, with a sharp decrease only at the location of the resistor.



4.7 Circuit Classification

Circuits may be classified according to whether they are linear or non-linear, and whether they contain active or passive elements. A linear circuit element has a linear (straight-line) relationship between voltage and current, meaning $V \propto I$; a non-linear element does not.

Definition 20 (Passive Linear Circuit). *Contains only passive linear elements, such as resistors, capacitors, inductors, and transformers.*

Definition 21 (Active Linear Circuit). *Contains at least one active element (e.g. transistor, op-amp, or independent source) while maintaining an overall linear $V-I$ relationship.*

Definition 22 (Passive Non-linear Circuit). *Contains only passive components but includes at least one non-linear element, such as a diode.*

Definition 23 (Active Non-linear Circuit). *Contains at least one active element and at least one non-linear element, such as a transistor with a non-linear $V-I$ characteristic like a diode.*

5 Circuit Components

5.1 Resistors

Definition 24. *A resistor is a device that opposes the flow of current through it, resulting in a voltage drop across it. Resistors are characterized by their resistance value R , measured in ohms (Ω).*

Proposition 10 (Series Resistors). *When resistors are connected in series, their resistances add: $R_{tot} = R_1 + R_2 + \dots + R_n$.*

Proposition 11 (Parallel Resistors). *When resistors are in parallel, they add inversely: $\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$.*

Example (Parallel Resistors). *Here is an illustration of different weird combination of resistors in parallel.*

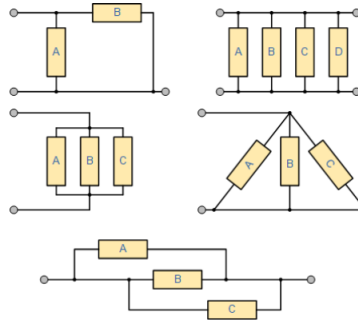


Figure 6: Examples of components wired in parallel.

5.1.1 Temperature Dependence

The resistance of many metals varies with temperature as

$$R_T = R_0 [1 + \alpha(T - T_0)],$$

where R_0 is the resistance at reference temperature T_0 and α is the linear temperature coefficient.

5.1.2 Power Dissipation

When current flows through a resistor, electrical energy is converted into heat. The power P dissipated is

$$P = \Delta V \cdot I = I^2 R = \frac{(\Delta V)^2}{R}.$$

The total energy dissipated over time is

$$U = \int R I^2(t) dt.$$

5.2 Batteries

Definition 25 (Battery). *A battery converts stored chemical energy into electrical energy by maintaining a potential difference between its terminals. Inside the battery, chemical reactions separate charge, creating an excess of negative charge at the anode and positive charge at the cathode.*

Definition 26 (Ideal Battery). *An ideal battery is a voltage source with zero internal resistance that maintains a constant voltage regardless of the current drawn.*

Definition 27 (Real Battery). *A real battery has internal resistance r that causes the terminal voltage to decrease as the current increases, leading to energy loss as heat.*

A real battery is modelled as an ideal voltage source V_0 in series with an internal resistance r . The terminal voltage when current I is drawn is

$$V_{\text{terminal}} = V_0 - Ir.$$

If an external resistor R is connected, Ohm's Law gives

$$V_0 = Ir + IR.$$

5.3 Ground and Reference

Definition 28 (Circuit Ground). *The reference point in a circuit at which the electric potential is defined to be zero. All voltages in the circuit are measured relative to this point. It does not have to be physically connected to the Earth.*

Symbol: $\equiv$.

Definition 29 (Chassis Ground). *A conductive frame or enclosure of a device used as a common return path for current; often connected to circuit ground. It provides shielding and safety by preventing exposed metal parts from reaching*

dangerous voltages. Symbol: ∇ .

Definition 30 (Earth Ground). *A physical connection to the Earth, typically made using a metal rod or pipe driven into the soil, used primarily for safety to allow fault currents to safely dissipate.*

5.4 Capacitors

Definition 31 (Capacitor). *A capacitor is a passive electrical component that stores energy in an electric field between two conductive plates separated by an insulating material (dielectric) or space.*

Proposition 12 (Capacitance). *The capacitance C is defined as the ratio of the charge Q stored on one plate to the voltage V across the plates:*

$$C = \frac{Q}{V}. \quad (2)$$

The SI unit of capacitance is the farad (F).

For a parallel plate capacitor,

$$C = \frac{\varepsilon A}{d},$$

where ε is the dielectric constant, A is the area of the plates, and d is the distance between them. The permittivity of free space is

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ F/m}.$$

Proposition 13 (Series Capacitors). *When capacitors are connected in series, they add inversely: $\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$.*

Proposition 14 (Parallel Capacitors). *When capacitors are connected in parallel, their capacitances add: $C_{tot} = C_1 + C_2 + \dots + C_n$.*

Remark 4. *Capacitors combine oppositely to resistors. This is because the voltage across capacitors in series is divided among them, while the charge on capacitors in parallel is shared.*

5.4.1 Dielectric Insulator

Definition 32 (Dielectric Insulator). *A dielectric is an insulating material that permits an electric field but does not conduct charge. When placed between capacitor plates, it increases capacitance by reducing the effective electric field, allowing more charge to be stored for a given voltage. Its permittivity is*

$$\varepsilon = \varepsilon_r \varepsilon_0,$$

where ε_r is the relative permittivity.

5.4.2 Energy Stored in a Capacitor

The energy stored equals the work required to move charge from one plate to the other. Since $V = Q/C$,

$$W = \int_0^Q V dQ = \int_0^Q \frac{Q}{C} dQ = \frac{1}{2} CV^2.$$

5.4.3 Charging a Capacitor

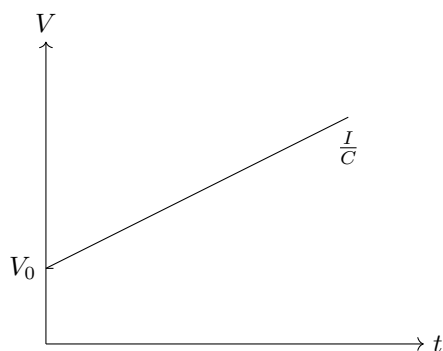
From (2) and (1), the current-voltage relationship for a capacitor is

$$I = C \frac{dV}{dt}, \quad V(t) = \frac{1}{C} \int I dt + V_0.$$

For constant current,

$$V(t) = V_0 + \frac{I}{C} t,$$

so the voltage increases linearly with slope I/C .



5.5 Inductors

Definition 33 (Inductor). *An inductor is a passive electrical component that stores energy in a magnetic field created by the flow of current through a coil of wire. The inductance L of an inductor is defined as the ratio of the magnetic flux Φ linking the coil to the current I flowing through it:*

$$L = \frac{\Phi}{I}.$$

The SI unit of inductance is the henry (H).

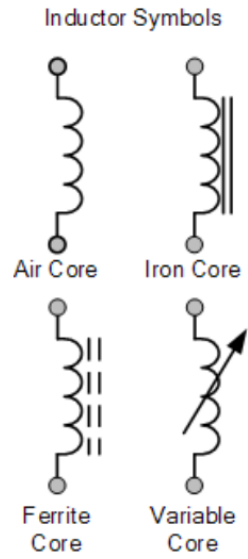
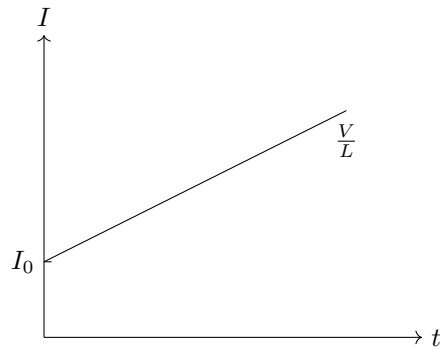


Figure 7: Circuit symbols for inductors with different core types.

5.5.1 Energy Storage

Energy is stored in the magnetic field within the inductor coil. For a constant applied voltage, the current in the coil rises linearly with slope V/L :



The energy stored is

$$W = \int_0^t V \cdot i \, dt = \int L \frac{di}{dt} \cdot i \, dt = \frac{1}{2} LI^2.$$

5.5.2 Networks of Inductors

Inductors in Series. The equivalent inductance is

$$L_{\text{total}} = \sum_{i=1}^N L_i.$$

Inductors in Parallel. The equivalent inductance satisfies

$$\frac{1}{L_{\text{total}}} = \sum_{i=1}^N \frac{1}{L_i}.$$

These relations are formally identical to those of resistors, but arise from the inductor voltage-current relation.

5.5.3 Solenoid

Definition 34 (Solenoid). *A solenoid is a long cylindrical coil of wire designed to produce a nearly uniform magnetic field inside it when current flows.*

For a solenoid of length l , cross-sectional area $A = \pi r^2$, number of turns N , and core permeability μ ,

$$L = \frac{\mu N^2 A}{l}.$$

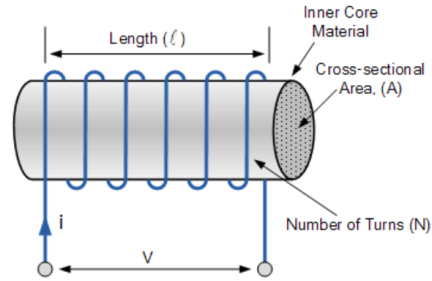
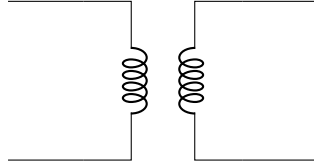


Figure 8: Schematic representation of a solenoid.

5.6 Transformers & Mutual Inductance (Not Tested)

Definition 35 (Transformer). *A transformer is a device consisting of two coils of wire (primary and secondary) magnetically coupled through a shared core. Its main purpose is to transfer energy between circuits to change voltage levels. Symbol:*



A transformer has N_1 and N_2 turns wound around a common core. A time-varying current I_1 in the primary produces flux Φ_B through the secondary. By Faraday's law,

$$\mathcal{E}_2 = -\frac{d\Phi_B}{dt} = -M\frac{dI_1}{dt}.$$

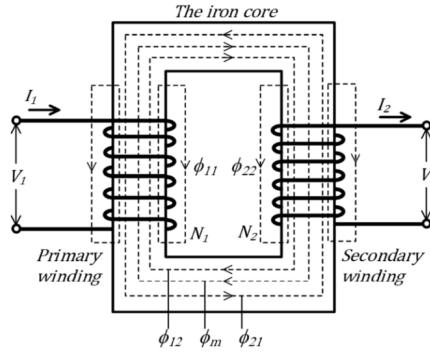


Figure 9: Visual representation of a transformer.

The SI unit of mutual inductance is the henry (H). If $I_1(t) = I_0 \sin(\omega t)$, then

$$V_2(t) = M\omega I_0 \cos(\omega t),$$

so the secondary voltage is sinusoidal but shifted in phase by $\pi/2$ relative to the current.

For an ideal transformer, power is conserved ($V_1 I_1 = V_2 I_2$), giving the transformer relations:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}.$$

Remark 5. Long-distance power lines use high-voltage AC because it significantly reduces energy loss during transmission: higher voltage means lower current, which minimises resistive heat loss. Transformers efficiently step AC voltage up and down as needed.

6 Circuit Analysis

6.1 RC Circuit

Definition 36 (RC Circuit). *An RC circuit consists of a resistor R , a capacitor C , and a constant source V_0 .*

Example (RC Circuit). *Visual representation of an RC circuit:*

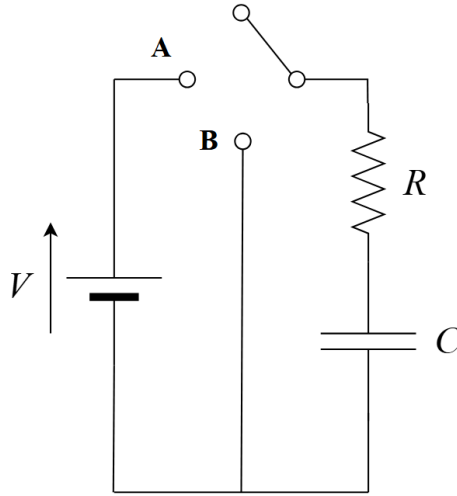


Figure 10: Visual representation of an RC circuit.

6.1.1 DC Analysis

Charging

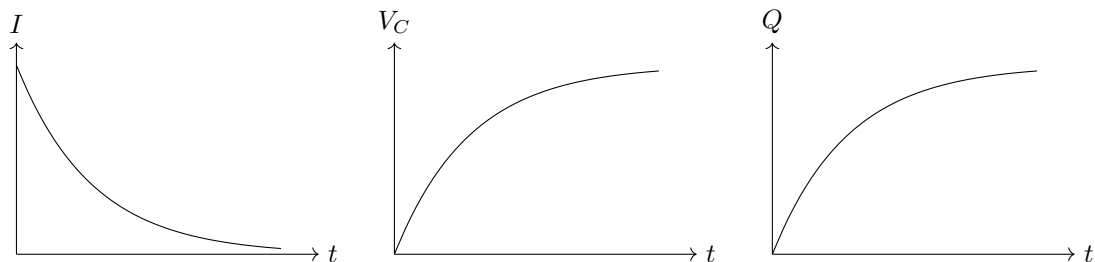
Consider an initially uncharged capacitor connected in series with a resistor and a constant source V_0 . Applying KVL,

$$V_0 = v_R(t) + v_C(t) = i(t)R + \frac{1}{C} \int_0^t i(\tau) d\tau.$$

Differentiating and solving,

$$I(t) = I_0 e^{-t/\tau} = \frac{V_0}{R} e^{-t/\tau}, \quad V_C(t) = V_0 (1 - e^{-t/\tau}),$$

where $\tau = RC$ is the time constant. As $t \rightarrow \infty$, the current decays to zero and $V_C \rightarrow V_0$. At $t = 0$, the capacitor behaves like a short circuit; as $t \rightarrow \infty$, it behaves like an open circuit.

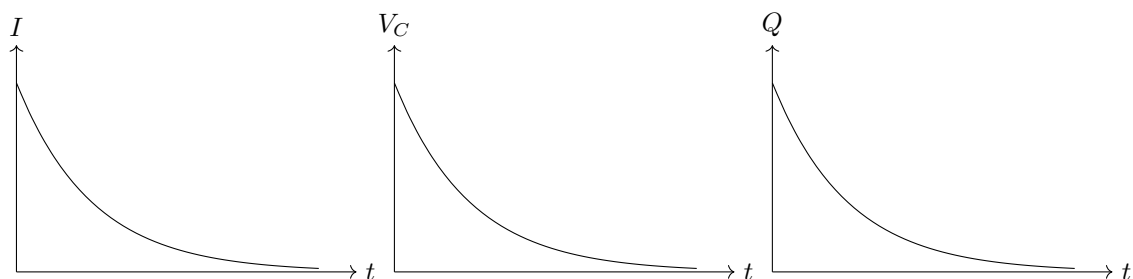


Discharging

If the source is removed and the capacitor initially has voltage V_0 :

$$I(t) = -\frac{V_0}{R} e^{-t/\tau}, \quad V_C(t) = V_0 e^{-t/\tau}.$$

Both current and voltage decay exponentially to zero.



6.1.2 AC Analysis

January 29,
2026

Let an RC circuit be driven by a periodic input with period T and an angular frequency $\omega = \frac{2\pi}{T} = 2\pi\nu$.

Definition 37 (Time Constant). *The time constant for an RC circuit is*

$$\tau_{RC} = RC.$$

Proposition 15 (Frequency Response). *The circuit behaviour depends on the ratio of T to τ :*

- *Low frequency:* $T \gg \tau$
- *High frequency:* $T \ll \tau$

If the input voltage is a square wave, the circuit response will differ significantly between low and high frequencies.

Proposition 16 (Low Frequency). *Consider a low-frequency ($T \gg \tau$) square-wave input, where the voltage alternates between $+V_0$ and $-V_0$ every half-cycle. The capacitor has enough time to fully charge and discharge during each half-cycle, resulting in the following behaviour:*

- Capacitor fully charges and discharges each half-cycle.
- Voltage drops almost entirely across the capacitor.
- Current is nearly zero (except for spikes at each voltage jump).
- Capacitor behaves approximately like an open circuit (break).

At low frequency, the capacitor has time to fully charge, so its voltage tracks the input: $\Delta V_C(t) \approx V(t)$. The current is therefore $I \approx C \frac{dV}{dt}$, and the voltage across the resistor is

$$\Delta V_R(t) = R \frac{dQ}{dt} \approx \tau \frac{dV_{\text{in}}(t)}{dt}, \quad \tau = RC.$$

The resistor acts as a differentiator. Sharp transitions in the square-wave input produce current spikes.

Proposition 17 (High Frequency). *For a high-frequency ($T \ll \tau$) square-wave input, the capacitor does not have time to charge significantly during each half-cycle, leading to the following behaviour:*

- Capacitor does not have time to charge.
- Very little voltage is dropped across the capacitor.
- Current values are close to what we expect for a pure resistor circuit; voltage profile is triangle-wave-like.
- Capacitor behaves approximately like a short circuit.

At high frequency, the capacitor has no time to charge, so the resistor voltage tracks the input: $\Delta V_R(t) \approx V(t)$, giving $I \approx V(t)/R$. The capacitor voltage is then

$$\Delta V_C(t) = \frac{Q}{C} \approx \frac{1}{\tau} \int_0^t V_{\text{in}}(t') dt', \quad \tau = RC.$$

The capacitor acts as an integrator. A square wave input produces a triangle-wave-like voltage across the capacitor.

6.2 LR Circuit

Definition 38 (LR Circuit). *An LR circuit is a circuit in which an inductor is connected in series with a resistor.*

Example (LR Circuit). Consider this circuit at the moment it is turned on (switch connects to ‘A’). If the inductor were replaced with a short, a current V/R would flow immediately, with all of the voltage drop across R .

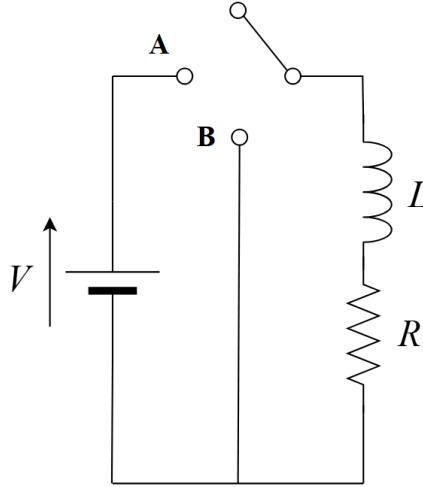


Figure 11: Visual representation of an LR circuit.

The current flow produces a magnetic flux through the coil. By Lenz’s law, the coil induces a voltage equal to the battery’s voltage, opposing the change in flux. Therefore, just at the instant the switch is closed, the current flow is zero.

For a single coil, the voltage-current relationship is

$$V = L \frac{di}{dt},$$

where L is the self-inductance.

6.2.1 DC Analysis

Charging

When the switch is connected to terminal A,

$$I(t) = \frac{V_0}{R} \left(1 - e^{-t/\tau}\right), \quad V_L(t) = V_0 e^{-t/\tau}.$$

At $t = 0$, an inductor behaves like an open circuit. As $t \rightarrow \infty$, it behaves like a short circuit and carries the full steady-state current V/R .

Discharging

$$I(t) = I_0 e^{-t/\tau}, \quad V_L(t) = V_0 e^{-t/\tau}.$$

Both current and voltage decay exponentially to zero.

6.2.2 AC Analysis

Let an LR circuit be driven by a periodic input with period T and angular frequency $\omega = \frac{2\pi}{T} = 2\pi\nu$.

Definition 39 (Time Constant). *The time constant for an LR circuit is*

$$\tau_{LR} = \frac{L}{R}.$$

Proposition 18 (Frequency Response). *The behavior of the circuit depends on the ratio of T to τ :*

- *Low frequency: $T \gg \tau$*
- *High frequency: $T \ll \tau$*

If the input voltage is a square wave, the circuit response will differ significantly between low and high frequencies.

Proposition 19 (Low Frequency). *Consider a low-frequency ($T \gg \tau$) square-wave input, where the voltage alternates between $+V_0$ and $-V_0$ every half-cycle. The inductor has enough time to fully respond during each half-cycle, resulting in the following behaviour:*

- *Inductor fully responds each half-cycle.*
- *Voltage drops almost entirely across the resistor.*
- *Current closely follows the input and is approximately square-wave-like.*
- *Inductor behaves approximately like a short circuit.*

At low frequency, the inductor has time to fully respond, so the resistor voltage tracks the input: $\Delta V_R(t) \approx V(t)$, giving $I \approx V_{\text{in}}(t)/R$. The voltage across the inductor is then

$$\Delta V_L(t) = L \frac{dI}{dt} \approx \tau \frac{dV_{\text{in}}(t)}{dt}, \quad \tau = \frac{L}{R}.$$

The inductor acts as a differentiator. Sharp transitions in the square-wave input produce voltage spikes across the inductor.

Proposition 20 (High Frequency). *For a high-frequency ($T \ll \tau$) square-wave input, the inductor does not have time to respond during each half-cycle, leading to the following behaviour:*

- Inductor does not have time to respond.
- Voltage drops almost entirely across the inductor.
- Current changes very little; the resistor voltage profile consists of sharp spikes at each transition.
- Inductor behaves approximately like an open circuit.

At high frequency, the inductor has no time to respond, so its voltage tracks the input: $\Delta V_L(t) \approx V(t)$, giving $L \frac{dI}{dt} \approx V_{in}(t)$. The voltage across the resistor is then

$$\Delta V_R(t) = IR \approx \frac{1}{\tau} \int_0^t V_{in}(t') dt', \quad \tau = \frac{L}{R}.$$

The resistor acts as an integrator. A square-wave input produces a triangle-wave-like voltage across the resistor.

6.3 Frequency Behaviour Summary

Circuit	Low frequency ($T \gg \tau$)	High frequency ($T \ll \tau$)
RC	Capacitor $\approx$ open (break)	Capacitor $\approx$ short
LR	Inductor $\approx$ short	Inductor $\approx$ open

Remark 6. A capacitor blocks DC and resists changes in voltage.

Remark 7. An inductor passes DC (after long time) and resists changes in current.

The LR circuit is the dual of the RC circuit. In an LR circuit, current plays the role that voltage plays in an RC circuit, and the integrator/differentiator roles of the components are swapped:

	RC Circuit	LR Circuit
Differentiator	Resistor (low freq.)	Inductor (low freq.)
Integrator	Capacitor (high freq.)	Resistor (high freq.)

7 Phasors and Complex Impedance

7.1 Complex Representation of AC Signals

Definition 40 (Complex Signal). A complex signal is not physical, but it embeds the real-valued physical signal $V(t)$ as its real part. It is a mathematical tool that simplifies the analysis of AC circuits by allowing us to represent sinusoidal signals as complex exponentials, which are easier to manipulate using algebraic methods.

Proposition 21 (Complex Signal Representation). A complex signal is defined as

$$\tilde{V}(t) = V_0 e^{j\phi(t)} = V_0 e^{j(\omega t + \phi_0)}.$$

Proposition 22 (Physical Signal from Complex Signal). *The real-valued physical signal $V(t)$ is the real part of the complex signal:*

$$V(t) = \operatorname{Re}\left[V_0 e^{j\phi(t)}\right] = \operatorname{Re}\left[\tilde{V}(t)\right].$$

7.2 Working with Complex Signals

To analyze an AC circuit using complex signals, perform circuit analysis⁴ with $\tilde{V}(t)$ to obtain a complex output signal $\tilde{V}_{\text{out}}(t)$. The real-valued physical output is then recovered by taking the real part:

$$V_{\text{out}}(t) = \operatorname{Re}\left[\tilde{V}_{\text{out}}(t)\right].$$

Alternatively, one can extract directly the change in signal amplitude and phase shift between input and output.

7.3 Capacitor Current

Consider a voltage signal $V(t) = \operatorname{Re}[\tilde{V}(t)]$ dropped across a capacitor. To find the current $I(t)$, we first compute the complex current $\tilde{I}(t)$ arising from $\tilde{V}(t)$, then take the real part:

$$\tilde{I}(t) = \frac{d\tilde{Q}}{dt} = C \frac{d\tilde{V}(t)}{dt} = jC\omega V_0 e^{j(\omega t + \phi_0)} = jC\omega \tilde{V}.$$

Extracting $I(t)$ using the fact that $j = e^{j\pi/2}$:

$$I(t) = \operatorname{Re}\left[\tilde{I}(t)\right] = C\omega V_0 \operatorname{Re}\left[e^{j(\omega t + \phi_0 + \pi/2)}\right] = C\omega V_0 \cos(\omega t + \phi_0 + \frac{\pi}{2}).$$

Proposition 23. *From the expression above, the current leads the input voltage by $\pi/2$; equivalently, the input voltage lags the current by $\pi/2$.*

Remark 8. *Note that I vanishes in the limit $\omega \rightarrow 0$ (DC limit), since in steady state the capacitor fully charges and fully opposes the voltage source.*

7.4 Impedance

Definition 41 (Impedance). *The impedance Z of a circuit element is a complex resistance defined as the ratio of the complex voltage across the element to the complex current through it:*

$$Z = \frac{\tilde{V}}{\tilde{I}}.$$

*The real part of Z governs energy dissipation; the imaginary part (the **reactance**) encodes the phase offset between voltage and current.*

⁴Circuit analysis is the process of determining the voltages and currents throughout a circuit by applying circuit laws (KVL, KCL) and component relationships. For AC circuits, this is done using the complex signal $\tilde{V}(t)$ in place of the real signal, which simplifies differentiation and integration into algebraic operations.

Proposition 24 (Impedance Combination). *Impedances combine exactly as resistances do:*

$$Z_{tot} = Z_1 + Z_2 \quad (\text{series}), \quad Z_{tot} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} \quad (\text{parallel}).$$

Definition 42 (Resistive Impedance). *The impedance of a resistor is purely real:*

$$Z_R = R.$$

Since Z_R has no imaginary part, a resistor introduces no phase offset between voltage and current.

Definition 43 (Capacitive Impedance). *The impedance of a capacitor is*

$$Z_C = \frac{1}{j\omega C}.$$

Since Z_C is purely imaginary and negative, the current leads the voltage by $\pi/2$. The magnitude $|Z_C| = 1/(\omega C)$ is large at low frequencies and falls off as ω increases; in the DC limit $\omega \rightarrow 0$, $Z_C \rightarrow \infty$, meaning the capacitor blocks DC signals entirely.

Remark 9. A capacitor causes current to lead voltage by $\pi/2$ because the current flowing into the capacitor must first accumulate charge on its plates before a voltage difference can be established. Since $I = C \frac{dV}{dt}$, maximum current flows when the voltage is changing fastest (crossing zero) and zero current flows when the voltage is stable at its peak.

Definition 44 (Inductive Impedance). *The impedance of an inductor is*

$$Z_L = j\omega L.$$

Since Z_L is purely imaginary and positive, the current lags the voltage by $\pi/2$. The magnitude $|Z_L| = \omega L$ grows with frequency, so the inductor increasingly opposes high-frequency signals.

Remark 10. An inductor causes current to lag voltage because its magnetic properties, governed by Lenz's law, oppose changes in current flow ($V = L \frac{dI}{dt}$), causing the current waveform to reach its maximum 90 after the voltage waveform. Voltage is proportional to the rate of change of current, meaning maximum voltage occurs when current changes fastest (at zero crossing) and zero voltage occurs when the current is constant (at its peak), forcing a 90 phase shift.

Proposition 25. *For any element with impedance Z , the complex current and voltage satisfy*

$$\tilde{I} = \frac{\tilde{V}}{Z},$$

the direct complex generalisation of Ohm's Law.

7.5 Power Dissipation of AC Signals

The power dissipated by a circuit element driven by an AC signal is found by averaging over one cycle:

$$P = \frac{1}{T} \int_0^T I(t) \Delta V(t) dt.$$

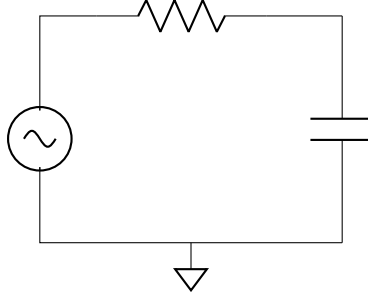
Taking $V(t) = V_0 \cos(\omega t)$ and $I(t) = I_0 \cos(\omega t + \phi_0)$, this gives

$$P = \frac{V_0 I_0}{T} \int_0^T \cos(\omega t) \cos(\omega t + \phi_0) dt = \frac{V_0 I_0}{2} \cos(\phi_0) = V_{\text{rms}} I_{\text{rms}} \cos(\phi_0).$$

For a capacitor and an inductor, $\phi_0 = \pm\pi/2$, so $P = 0$; they do not dissipate energy but instead store and release it. For a resistor, $\phi_0 = 0$, so $P = V_{\text{rms}} I_{\text{rms}}$; all the energy is dissipated as heat.

7.6 Sinusoidally Driven RC Circuit

Example (Current in Sinusoidally Driven RC Circuit). Let $\tilde{V}(t) = V_0 e^{j\omega t}$, $Z_R = R$, and $Z_C = \frac{1}{j\omega C}$, find the current in the following RC circuit:



First and foremost, we need to find the total impedance,

$$Z_{\text{tot}} = Z_R + Z_C = R + \frac{1}{j\omega C} = \frac{1 + j\omega RC}{j\omega C} = \frac{1 + j\omega\tau}{j\omega C},$$

where $\tau = RC$.

After finding the total impedance, we are able to find the complex current,

$$\tilde{I}(t) = \frac{\tilde{V}(t)}{Z_{\text{tot}}} = \frac{j\omega C V_0 e^{j\omega t}}{1 + j\omega\tau}.$$

The next step is to find the magnitude of the total impedance

$$|Z_{\text{tot}}|^2 = R^2 + \frac{1}{\omega^2 C^2} = R^2 \frac{1 + (\omega\tau)^2}{(\omega\tau)^2},$$

for which we get the amplitude of the current

$$|\tilde{I}| = \frac{V_0}{|Z_{\text{tot}}|} = \frac{V_0}{R} \cdot \frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}}.$$

Remark 11. At $\omega = 0$ (DC limit) the current vanishes; as $\omega \rightarrow \infty$ it rises to $I_{\max} = V_0/R$, since the capacitive impedance vanishes.

For the last step, we need to find the phase offset, since this is an RC circuit, the current leads the voltage due to the capacitor by

$$\phi = \arctan\left(\frac{1}{\omega\tau}\right).$$

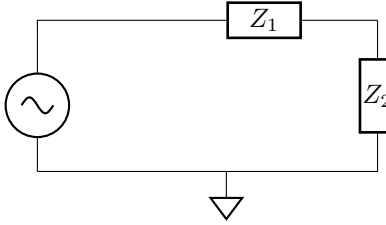
The full complex current is therefore

$$\tilde{I}(t) = \frac{V_0}{R} \cdot \frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}} e^{j(\omega t + \phi)}.$$

Remark 12 (Limiting Cases). In the DC limit ($\omega \rightarrow 0$), the voltage drops almost entirely across the capacitor and the current leads the voltage by $\phi = \arctan(\infty) = \pi/2$. In the high-frequency limit ($\omega \rightarrow \infty$), the voltage drops almost entirely across the resistor and the current is in phase with the voltage: $\phi = \arctan(0) = 0$.

7.7 Transfer Function

Consider a general voltage divider circuit built from two impedances Z_1 and Z_2 in series, driven by a sinusoidal source. The output voltage $\tilde{V}'$ is taken across Z_2 :



The transfer function is therefore

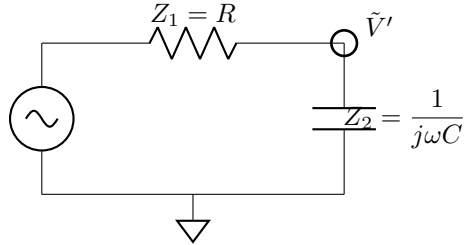
$$\frac{\tilde{V}'}{\tilde{V}} = \frac{Z_2}{Z_1 + Z_2} = \left| \frac{\tilde{V}'}{\tilde{V}} \right| e^{j\phi},$$

where $\left| \frac{\tilde{V}'}{\tilde{V}} \right|$ describes how the output amplitude is affected relative to the input amplitude, and ϕ is the phase shift of the output relative to the input.

7.8 Filters and Bode Plots

Definition 45 (Bode Plot). A Bode plot shows a circuit's frequency response: amplitude ratio (in dB) and phase shift (in degrees) versus frequency on a logarithmic scale, where $V_{dB} = 20 \log_{10}(V)$.

Example (RC Low-Pass Filter). Consider the following circuit, with output taken across the capacitor:



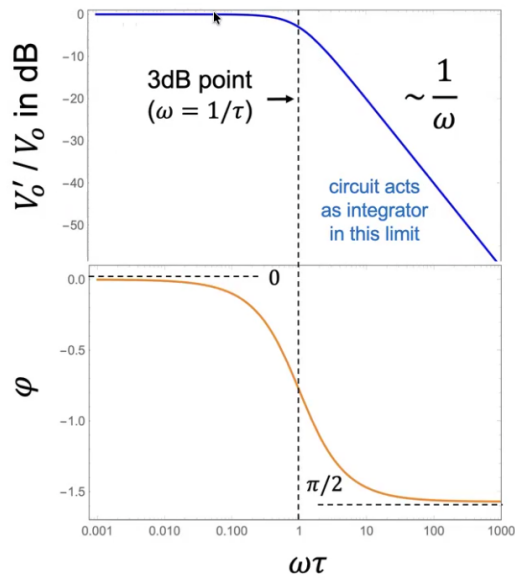
Output across the capacitor gives transfer function

$$\frac{\tilde{V}'}{\tilde{V}} = \frac{1}{1 + j\omega\tau} = \frac{1}{1 + (\omega\tau)^2} - j \frac{\omega\tau}{1 + (\omega\tau)^2}, \quad \tau = RC,$$

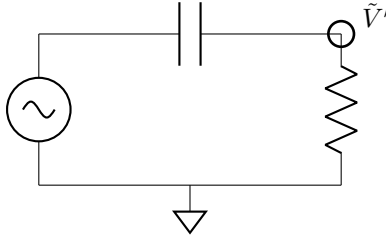
with amplitude and phase

$$\left| \frac{\tilde{V}'}{\tilde{V}} \right|^2 = \frac{1}{1 + (\omega\tau)^2}, \quad \phi = -\arctan(\omega\tau).$$

At high frequencies ($\omega\tau \gg 1$) the amplitude falls as $1/\omega\tau$ and the circuit acts as an integrator. Bode plots (linear then log scale):



Example (RC High-Pass Filter). Consider the following circuit, with output taken across the resistor:

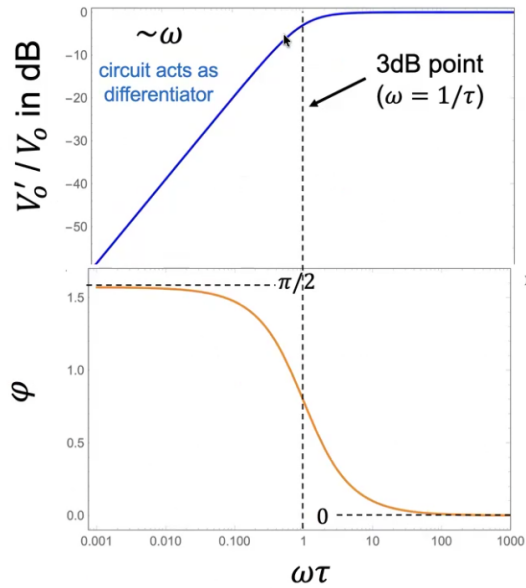


Output across the resistor gives

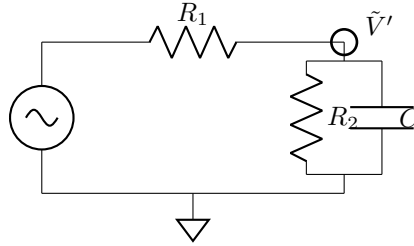
$$\frac{\tilde{V}'}{\tilde{V}} = \frac{j\omega\tau}{1 + j\omega\tau} = \frac{(\omega\tau)^2}{1 + (\omega\tau)^2} + j \frac{\omega\tau}{1 + (\omega\tau)^2},$$

with amplitude and phase

$$\left| \frac{\tilde{V}'}{\tilde{V}} \right|^2 = \frac{(\omega\tau)^2}{1 + (\omega\tau)^2}, \quad \phi = \arctan\left(\frac{1}{\omega\tau}\right).$$



Example (Two-Stage RC Circuit). Consider the following circuit, in which a resistor R_2 is placed in parallel with the capacitor:



With $Z_1 = R_1$ and $Z_2 = R_2 \parallel \frac{1}{j\omega C}$, the transfer function is

$$\frac{\tilde{V}'}{\tilde{V}} = \frac{1}{R_1 \left(\frac{1}{R_2} + j\omega C \right) + 1}.$$

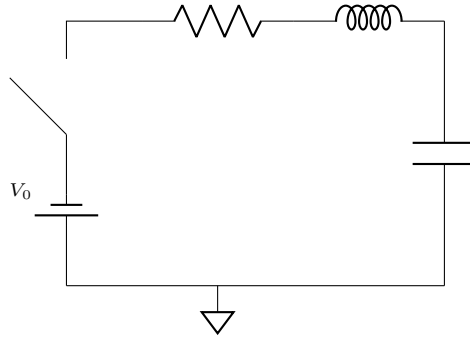
Limiting cases: as $\omega \rightarrow \infty$ the capacitor shorts R_2 , recovering a low-pass RC ($\tilde{V}'/\tilde{V} \rightarrow 1/(1 + j\omega R_1 C)$); as $\omega \rightarrow 0$ the capacitor opens, giving a resistive divider ($\tilde{V}'/\tilde{V} \rightarrow R_2/(R_1 + R_2)$).

8 RLC Circuits

February 17,
2026

8.1 Charging with a Step Voltage

Consider the following series RLC circuit, with a switch closed at $t = 0$ to create a voltage step:



Summing voltage drops around the loop (KVL) gives

$$V_0 = RI + L \frac{dI}{dt} + \frac{Q}{C}.$$

Differentiating with respect to time and dividing through by L ,

$$\frac{d^2 I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \frac{1}{LC} I = 0.$$

We define the natural resonant frequency

$$\omega_0 \equiv \frac{1}{\sqrt{LC}},$$

at which the inductive and capacitive reactances are equal in magnitude. At this frequency, energy oscillates efficiently between the inductor and capacitor, the circuit is purely resistive, and the impedance is minimised. We also define the damping factor

$$\Gamma \equiv \frac{R}{L},$$

which represents the rate at which energy is dissipated by the resistor. The equation of motion becomes

$$\frac{d^2 I}{dt^2} + \Gamma \frac{dI}{dt} + \omega_0^2 I = 0.$$

8.2 Solution by Characteristic Equation

Substituting the trial solution $I(t) = ce^{pt}$ gives the characteristic equation

$$p^2 + \Gamma p + \omega_0^2 = 0 \implies p = -\frac{\Gamma}{2} \pm \sqrt{\left(\frac{\Gamma}{2}\right)^2 - \omega_0^2}.$$

The nature of the solution depends on the sign of the discriminant, yielding three distinct cases.

8.3 Damping Cases

Case	Condition	Behaviour
Underdamped	$\Gamma < 2\omega_0$	Oscillatory, decaying envelope
Critically damped	$\Gamma = 2\omega_0$	Fastest non-oscillatory return
Overdamped	$\Gamma > 2\omega_0$	Slow, non-oscillatory decay

8.3.1 Underdamped ($\Gamma < 2\omega_0$).

The discriminant is negative, giving complex roots. The solution is a sinusoid with exponentially decaying envelope:

$$I(t) = \frac{V_0}{L\omega_f} \sin(\omega_f t) e^{-\Gamma t/2},$$

where the damped frequency is

$$\omega_f = \sqrt{\omega_0^2 - \left(\frac{\Gamma}{2}\right)^2}.$$

This is the most physically interesting case, exhibiting oscillatory behaviour before settling. As $R \rightarrow 0$, damping vanishes, $\omega_f \rightarrow \omega_0$, and in the unphysical limit of a pure LC circuit the oscillation persists indefinitely at the natural frequency ω_0 .

8.3.2 Overdamped ($\Gamma > 2\omega_0$).

The discriminant is positive, giving two distinct real roots. The solution decays without oscillation:

$$I(t) = \frac{V_0}{L\omega'_f} \sinh(\omega'_f t) e^{-\Gamma t/2},$$

where

$$\omega'_f = \sqrt{\left(\frac{\Gamma}{2}\right)^2 - \omega_0^2}.$$

8.3.3 Critically damped ($\Gamma = 2\omega_0$).

The discriminant is zero, giving a repeated real root. The solution is

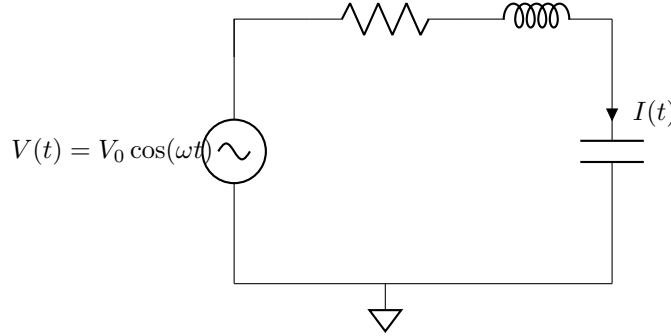
$$I(t) = \frac{V_0}{L} t e^{-\Gamma t/2}.$$

This case represents the fastest return to equilibrium without oscillation.

8.4 RLC Circuit with AC Driving

February 19,
2026

Consider the following series RLC circuit driven by $V(t) = V_0 \cos(\omega t)$:



With $\tilde{V} = V_0 e^{j\omega t}$, $\tau = L/R$, and $\omega_0 = 1/\sqrt{LC}$, the total impedance is $Z_{\text{tot}} = R + j\omega L + \frac{1}{j\omega C}$, giving complex current

$$\tilde{I} = \frac{\tilde{V}}{Z_{\text{tot}}} = \frac{\tilde{V}}{R} \cdot \frac{j\omega\tau}{1 - \frac{\omega^2}{\omega_0^2} + j\omega\tau}.$$

The amplitude of the current is

$$|\tilde{I}|^2 = \left(\frac{V_0}{R}\right)^2 \frac{(\omega\tau)^2}{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + (\omega\tau)^2}.$$

At $\omega = \omega_0$ the denominator reduces to $(\omega_0\tau)^2$, so $|\tilde{I}|$ reaches its maximum $I_{\max} = V_0/R$. At resonance, the capacitive and inductive impedances cancel exactly, leaving only R .

We can obtain the phase offset by multiplying through by the complex conjugate of the denominator,

$$\frac{\tilde{I}}{\tilde{V}/R} = \frac{(\omega\tau)^2 + j\omega\tau\left(1 - \frac{\omega^2}{\omega_0^2}\right)}{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + (\omega\tau)^2}, \quad \tan\phi = \frac{1 - \frac{\omega^2}{\omega_0^2}}{\omega\tau}.$$

The physical current is therefore $I(t) = |\tilde{I}| \cos(\omega t + \phi)$.

8.5 Quality Factor

Definition 46 (Quality Factor). *The quality factor Q is a dimensionless parameter that quantifies the sharpness of a resonance peak in a resonant system. A high Q corresponds to a narrow resonance.*

Proposition 26 (Quality Factor of an RLC Circuit). *The quality factor can be defined as*

$$Q = \frac{1}{\omega_0\tau} = \frac{\omega_0 L}{R}.$$

As $R \rightarrow 0$, the circuit approaches a pure LC resonator, $Q \rightarrow \infty$, and only oscillation at ω_0 can be sustained. At resonance the capacitive and inductive reactances are equal in magnitude but opposite in phase (offset by π radians), so they cancel exactly and the circuit presents purely resistive impedance to the driving source.

8.6 Power and Resonance

Proposition 27 (Instantaneous Power). *The instantaneous power delivered to the circuit is*

$$P(t) = V(t) I(t) = V_0 I_0 \cos(\omega t) \cos(\omega t + \phi).$$

Time-averaging over one complete cycle gives

$$\langle P \rangle = \frac{1}{T} \int_0^T P(t) dt = \frac{V_0 I_0}{2} \cos\phi.$$

Definition 47 (Power Factor). *The factor $\cos\phi$ from the previous equation is called the power factor.*

Remark 13. *Capacitors and inductors introduce a phase offset ϕ that reduces the efficiency of power delivery. In a pure LC circuit the phase offset is necessarily $\pm\pi/2$, so $\cos\phi = 0$ and the time-averaged power input vanishes — no net energy is delivered unless the circuit is driven exactly at resonance $\omega = \omega_0$, where the inductive and capacitive impedances cancel and $\phi = 0$.*

8.7 Capacitor Voltage in a Driven RLC Circuit

For the driven series RLC circuit of Section 8.4, the voltage drop across the capacitor is

$$\Delta V_C = \frac{Q}{C} = \frac{1}{C} \int_0^t I(t') dt'.$$

Using the complex current $\tilde{I} = |\tilde{I}|e^{j(\omega t + \phi)}$, integrating gives

$$\tilde{V}_C = \frac{\tilde{I}}{j\omega C} = \frac{V_0}{\sqrt{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + (\omega\tau)^2}} e^{j(\omega t + \phi - \pi/2)}.$$

Equivalently, from the impedance divider,

$$\frac{\tilde{V}'}{\tilde{V}} = \frac{Z_C}{Z_{\text{tot}}} = \frac{1}{1 - \frac{\omega^2}{\omega_0^2} + j\omega\tau},$$

with amplitude and phase

$$\left| \frac{\tilde{V}'}{\tilde{V}} \right|^2 = \frac{1}{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + (\omega\tau)^2}, \quad \phi = -\arctan\left(\frac{\omega\tau}{1 - \frac{\omega^2}{\omega_0^2}}\right).$$

The capacitor voltage peaks near, but slightly below, ω_0 , because the peak is determined by the combination of maximum current flow and the frequency-dependent capacitive reactance, rather than by maximising the current alone.

8.7.1 Limiting Behaviour

Low frequency ($\omega \ll \omega_0$). The capacitive impedance dominates and nearly all of the input voltage is dropped across the capacitor. The output amplitude approaches V_0 and the phase offset approaches zero.

At resonance ($\omega = \omega_0$). Current is limited only by R and reaches its maximum. Large voltage drops appear across both the inductor and capacitor, but they cancel. The current is in phase with $V(t)$; since the capacitor voltage lags the current by $\pi/2$, it lags the input by $\pi/2$.

High frequency ($\omega \gg \omega_0$). The voltage drops almost entirely across the inductor, so $V(t) \approx L \frac{dI}{dt}$, giving

$$I(t) \approx \frac{1}{L} \int V(t') dt' = \frac{V_0}{\omega L} \sin(\omega t).$$

The capacitor voltage is then

$$\Delta V_C = \frac{1}{C} \int I(t') dt' \approx -\frac{V_0}{\omega^2 LC} \cos(\omega t) = -\left(\frac{\omega_0}{\omega}\right)^2 V(t).$$

The inductor current lags $V(t)$ by $\pi/2$; the capacitor voltage lags the current by a further $\pi/2$, so the capacitor voltage lags $V(t)$ by π in total.

Remark 14. The **natural frequency** ω_0 is the inherent rate at which energy oscillates between the inductor and capacitor in the absence of external driving. The **resonant frequency** is the specific driving frequency that produces maximum current. For low damping these are nearly identical; for large resistance they diverge.

8.8 Quality Factor

The quality factor Q_f is a dimensionless measure of the ratio of energy stored to energy dissipated per cycle:

$$Q_f = 2\pi \frac{W_s}{W_L},$$

where W_s is the energy stored in the circuit and W_L is the energy lost as heat per cycle.

8.8.1 Derivation for a Series RLC Circuit

Taking $I(t) = I_0 \cos(\omega_0 t)$, the maximum energy stored (in the inductor at peak current) is

$$W_s = \frac{1}{2} L I_0^2.$$

By Joule's law⁵, the energy dissipated per cycle $T_0 = 2\pi/\omega_0$ is

$$W_L = R \int_0^{T_0} I_0^2 \cos^2(\omega_0 t) dt = \frac{R I_0^2 T_0}{2} = \frac{\pi R I_0^2}{\omega_0}.$$

Therefore,

$$Q_f = 2\pi \cdot \frac{\frac{1}{2} L I_0^2}{\frac{\pi R I_0^2}{\omega_0}} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 \tau}.$$

8.9 Quality Factor and Energy Decay

The quality factor is also directly related to the rate at which energy drains from a freely discharging RLC circuit. The current amplitude decays as

$$I_0(t) = I_0(0) e^{-\Gamma t/2}, \quad \Gamma = \frac{R}{L},$$

⁵Joule's heating law states that the heat produced in a conductor is proportional to the square of the current, the resistance, and the duration of flow.

so the stored energy decays as

$$E(t) = \frac{1}{2}LI_0^2(t) = E_0 e^{-\Gamma t} \implies \frac{dE}{dt} = -\Gamma E.$$

The number of radians of oscillation completed per energy e -folding time is

$$Q_f = \omega_0 \frac{E}{|dE/dt|} = \frac{\omega_0}{\Gamma} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 \tau},$$

consistent with the result above. A high Q_f therefore means the circuit loses a small fraction of its stored energy per oscillation cycle.

8.10 Power Input into a Driven RLC Series Circuit

The instantaneous power delivered to the circuit is $P(t) = V(t)I(t)$, where $V(t) = V_0 \cos(\omega t)$ and $I(t) = I_0(\omega) \cos(\omega t + \phi(\omega))$. Time-averaging over one cycle gives

$$\langle P(\omega) \rangle = \frac{1}{T} \int_0^T V(t) I(t) dt = \frac{V_0 I_0(\omega)}{2} \cos \phi(\omega) = \frac{V_0^2}{2R} \cdot \frac{(\omega \tau)^2}{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + (\omega \tau)^2}.$$

Substituting $x \equiv \omega/\omega_0$ and $Q_f = \omega_0 L/R = 1/(\omega_0 \tau)$, this becomes

$$\langle P(x) \rangle = P_{\max} \cdot \frac{\frac{x^2}{Q_f^2}}{(1 - x^2)^2 + \frac{x^2}{Q_f^2}}, \quad P_{\max} \equiv \frac{V_0^2}{2R}.$$

The power curves are normalised by $P_{\max}$, which itself depends on R and hence on Q_f :

- **High Q_f :** more power can be stored in the circuit, but only over a narrow range of driving frequencies.
- **Low Q_f :** less power is stored, but the circuit responds over a broader frequency range.

8.10.1 Steady State

A driven RLC circuit does not accumulate energy indefinitely. The power input from the driving source is balanced by the power dissipated in the resistor, leading to a stable steady-state energy level. While the inductor and capacitor continuously exchange energy, the resistor converts electrical energy to heat, preventing unlimited accumulation. In the idealised limit $R \rightarrow 0$, physical effects such as inductor saturation, capacitor breakdown, or wire melting would prevent infinite energy storage in practice.

8.10.2 Bandwidth and Quality Factor

Define the **bandwidth** $\Delta\omega = \omega_2 - \omega_1$ as the frequency interval between the two half-power points, i.e. the frequencies at which $\langle P \rangle = P_{\max}/2$. Then

$$Q_f = \frac{\omega_0}{\Delta\omega}.$$

This relation is a general property of resonators and is not specific to RLC circuits. In the context of radio receivers, the RLC circuit acts as a tuner: power delivered from incoming radio waves is maximised at the resonant frequency, while signals at other frequencies are suppressed. A high Q_f allows the receiver to distinguish closely spaced stations; a low Q_f gives a broader but less selective response.

9 Circuit Network Analysis

9.1 Kirchhoff's Laws (Review)

Definition 48 (Kirchhoff's Current Law (KCL)). *At any junction in a circuit, the sum of all currents entering equals the sum of all currents leaving.*

Definition 49 (Kirchhoff's Voltage Law (KVL)). *Around any closed loop in a circuit, the algebraic sum of all voltage changes equals zero.*

9.2 Terminology

Definition 50 (Node). *Any point in a circuit where two or more components connect.*

Definition 51 (Junction (Principal Node)). *A point where three or more branches meet. All junctions are nodes, but not all nodes are junctions. Nodes describe general connectivity; junctions are the key locations for analysing current division.*

Definition 52 (Branch). *A single path connecting two nodes, containing one or more circuit elements in series.*

9.3 Methods of Analysis

Kirchhoff's laws are straightforward to state but require systematic application in complex circuits. Three standard methods exist.

9.3.1 Branch Method

A current variable is assigned to each branch; KCL is written for each junction and KVL for each independent loop. If there are n unknown branch currents and k junctions, this produces $k - 1$ independent KCL equations and $n - k + 1$

independent KVL equations. This is the most general method but typically yields the largest system of simultaneous equations, making it inefficient for complex networks.

9.3.2 Loop Current Method

A single circulating current is assigned to each independent loop. KVL is applied to each loop; KCL is automatically satisfied. This method is best suited to circuits with many series elements or voltage sources. It can only be applied to **planar circuits**⁶.

9.3.3 Nodal Method

Voltage levels at each node are taken as the unknowns, with one node designated as the reference (ground). KCL is then applied at each non-reference node, expressing branch currents in terms of node voltages via Ohm's law (or its complex generalisation for AC circuits). The number of equations is reduced to the number of nodes minus one. This method is best suited to circuits with many parallel elements or current sources.

Method	Best suited for	Equations required
Branch	General circuits	n (largest system)
Loop	Many series elements / voltage sources	Independent loops only
Nodal	Many parallel elements / current sources	Nodes -1

9.4 Superposition Theorem

Theorem 3 (Superposition). *In any circuit containing only linear elements, the current at any point is equal to the algebraic sum of the currents produced by each independent source acting alone. To find the contribution of a single voltage source, all other voltage sources are replaced by short circuits (plain wires) and all other current sources are replaced by open circuits.*

9.5 Current Sources

Just as a voltage source maintains a fixed potential difference regardless of the current drawn, a **current source** supplies a fixed current regardless of the voltage across it.

Definition 53 (Ideal Current Source). *An ideal current source supplies a fixed current that is completely independent of the voltage across its terminals, corresponding to an internal resistance of $r = \infty$.*

⁶A planar circuit is one that can be drawn on a flat surface with no crossing wires. Non-planar circuits contain branches that cannot be laid out in a plane without intersections, and the loop method cannot be applied directly to them.

Definition 54 (Real Current Source). *A real current source has a finite internal resistance r , causing the delivered current to depend on the terminal voltage. It is modelled as an ideal current source in parallel with a resistance r .*

A current source can be realised in two ways:

- **Passive approximation:** Apply a voltage across a resistor whose resistance is much larger than the net resistance of the load to which the current is being supplied. The current through the load is then approximately constant at $I \approx V/r$.
- **Active implementation:** Use nonlinear active elements such as transistors to enforce a fixed current over a wide range of terminal voltages.

9.6 Converting Between Source Types

To create a constant voltage source from a constant current source, a component is placed in parallel that shunts (diverts) excess current while maintaining a fixed voltage drop. This requires the shunt resistance to satisfy $r \ll R_L$, so that the terminal voltage remains approximately constant regardless of the load.

9.7 Thévenin's Theorem

Theorem 4 (Thévenin's Theorem). *Any combination of voltage sources, current sources, and resistances with two terminals A and B is electrically equivalent to an ideal voltage source e in series with a resistance r .*

This theorem extends to circuits containing reactive elements (capacitors and inductors), in which case the equivalent circuit is an ideal AC voltage source in series with a complex impedance Z .

Since the Kirchhoff loop equations are linear, a linear relation must exist between the terminal voltage ΔV_{AB} and the current I_{AB} drawn at A and returned at B:

$$\Delta V_{AB} = e - I_{AB} r.$$

In analogy with a real battery, the constant e is the open-circuit voltage and r is the series resistance. The negative sign reflects the fact that drawing current increases the voltage drop across internal resistive elements, reducing ΔV_{AB} .

9.7.1 Determining e and r

Approach 1: Experimental (black-box circuit). Connect a variable load resistor R_L between terminals A and B and adjust it to vary I_{AB} . From the Thévenin relation,

$$I_{AB} = \frac{e}{r} - \frac{\Delta V_{AB}}{r}.$$

Alternatively, measure ΔV_{AB} as a function of R_L :

$$\Delta V_{AB} = e - r \frac{\Delta V_{AB}}{R_L} \implies \Delta V_{AB} = \frac{e R_L}{r + R_L} = \frac{e}{1 + \frac{r}{R_L}}.$$

Plotting ΔV_{AB} versus R_L allows both e and r to be extracted from a fit.

Approach 2: Theoretical (known circuit diagram).

- **Thévenin voltage e :** compute the open-circuit voltage between A and B with no current drawn ($I_{AB} = 0$).
- **Thévenin resistance r :** deactivate all independent sources (replace voltage sources with short circuits and current sources with open circuits), then compute the total resistance seen between terminals A and B.

9.8 Norton's Theorem

Theorem 5 (Norton's Theorem). *Any combination of voltage sources, current sources, and resistances with two terminals A and B is electrically equivalent to an ideal current source i in parallel with a resistance r_N .*

9.8.1 Equivalence of Thévenin and Norton Representations

The two theorems are equivalent. Applying KCL at the output node of the Norton circuit,

$$i = I_{AB} + \frac{\Delta V_{AB}}{r_N} \implies \Delta V_{AB} = r_N i - r_N I_{AB}.$$

Identifying $e \equiv r_N i$ and $r \equiv r_N$ recovers the Thévenin relation $\Delta V_{AB} = e - r I_{AB}$. The conversion relations between the two representations are therefore

$$e = r_N i, \quad r = r_N.$$

10 Fourier Analysis

10.1 Fourier Series

Periodic signals can be expressed as sums of sinusoidal components of varying frequency. Let $f(t)$ be periodic with period T , so that $f(t) = f(t + T)$.

Definition 55 (Fourier Series). *The Fourier series of $f(t)$ is its expansion as a sum of cosine and sine functions:*

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)],$$

where $\omega_n = \frac{2\pi n}{T}$. The term $n = 1$ is the **fundamental**, $n = 2$ is the **first harmonic**, $n = 3$ is the **second harmonic**, and so on. The constant $a_0/2$ captures the non-varying (DC) component of the signal.

Both sine and cosine terms are required in order to capture the phase of the periodic function being decomposed. Both families of functions are needed because neither alone forms a complete basis for arbitrary periodic signals.

Example. $f(t) = \cos^2(at) = \frac{1}{2} + \frac{1}{2} \cos(2at)$, so $a_0 = 1$, $a_1 = \frac{1}{2}$, with $a = \pi/T$.

Example.

$$\begin{aligned} f(t) = \cos^4(at) &= \left(\frac{1}{2} + \frac{1}{2} \cos(2at)\right)^2 \\ &= \frac{1}{4} + \frac{1}{2} \cos(2at) + \frac{1}{4} \cos^2(2at) \\ &= \frac{3}{8} + \frac{1}{2} \cos(2at) + \frac{1}{8} \cos(4at), \end{aligned}$$

so $a_0 = 3/4$, $a_1 = 1/2$, $a_2 = 1/8$, with $a = \pi/T$.

10.2 Orthogonality Relations

The cosine and sine functions satisfy the following orthogonality integrals, where δ_{nm} is the Kronecker delta:

$$\delta_{nm} = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases}.$$

1. $\int_{-T/2}^{T/2} \cos\left(\frac{2\pi n}{T}t\right) \cos\left(\frac{2\pi m}{T}t\right) dt = \frac{T}{2} \delta_{nm}$
2. $\int_{-T/2}^{T/2} \sin\left(\frac{2\pi n}{T}t\right) \sin\left(\frac{2\pi m}{T}t\right) dt = \frac{T}{2} \delta_{nm}$
3. $\int_{-T/2}^{T/2} \cos\left(\frac{2\pi n}{T}t\right) \sin\left(\frac{2\pi m}{T}t\right) dt = 0$

These relations express the fact that the cosines and sines form a **complete orthogonal basis** for periodic functions on $[-T/2, T/2]$.

10.3 Fourier Coefficients

Using the orthogonality relations, the Fourier coefficients are extracted by projection onto each basis function:

1. $a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(\omega_n t) dt$
2. $b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(\omega_n t) dt$

$$3. a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(t) dt$$

The constant a_0 is simply twice the time-average of $f(t)$ over one period.

10.4 Convergence: Dirichlet's Theorem

Theorem 6 (Dirichlet's Theorem). *The Fourier series converges to $f(t)$ at all points where $f(t)$ is continuous. At a jump discontinuity, the series converges to the midpoint of the jump.*

The theorem holds if and only if on the interval $[-T/2, T/2]$:

1. $f(t)$ is single-valued.
2. $f(t)$ has a finite number of maxima and minima.
3. $f(t)$ has a finite number of discontinuities.
4. $\int_{-T/2}^{T/2} |f(t)|^2 dt$ is finite.

Any physically realisable waveform satisfies these conditions.

10.5 Fourier Transform

The Fourier series applies to periodic signals. For aperiodic signals, the generalisation is the **Fourier transform**, which decomposes a signal into a continuous spectrum of frequencies.

Definition 56 (Fourier Transform).

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt.$$

Definition 57 (Inverse Fourier Transform).

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega.$$

Here $f(t)$ is the **time-domain** representation of the signal, and $F(\omega)$ is its **frequency-domain** representation, describing how the signal's energy is distributed across frequencies.

10.6 Gibbs Phenomenon

At a jump discontinuity, truncating the Fourier series at a finite number of terms N produces periodic overshoots and undershoots near the jump. These do not decrease in height as N increases — the overshoot remains approximately 9% of the jump — but the *width* of the oscillations shrinks to zero as $N \rightarrow \infty$. This is known as the **Gibbs phenomenon**.

Remark 15. *Gibbs phenomenon does not contradict Dirichlet's theorem. The theorem guarantees convergence to the midpoint of the jump in the limit $N \rightarrow \infty$; it makes no claim about the behaviour of partial sums at finite N .*

10.7 Example: Sawtooth Wave

Consider $f(t) = t/T$ for $|t| \leq T/2$. Since $f(t)$ is odd, $a_n = 0$ for all n . The sine coefficients are

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(\omega_n t) dt = \frac{2}{T^2} \int_{-T/2}^{T/2} t \sin\left(\frac{2\pi n t}{T}\right) dt = -\frac{(-1)^n}{n\pi}.$$

10.8 Even and Odd Functions

The parity of $f(t)$ simplifies the computation of Fourier coefficients, since the product of two functions of the same parity is even, and of opposite parity is odd. An integral of an odd function over a symmetric interval vanishes.

1. If $f(t)$ is **even** ($f(-t) = f(t)$):

$$(a) \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(\omega_n t) dt = \frac{4}{T} \int_0^{T/2} f(t) \cos(\omega_n t) dt \quad (\text{even} \times \text{even} = \text{even})$$

$$(b) \quad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(\omega_n t) dt = 0 \quad (\text{even} \times \text{odd} = \text{odd})$$

2. If $f(t)$ is **odd** ($f(-t) = -f(t)$):

$$(a) \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(\omega_n t) dt = 0 \quad (\text{odd} \times \text{even} = \text{odd})$$

$$(b) \quad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(\omega_n t) dt = \frac{4}{T} \int_0^{T/2} f(t) \sin(\omega_n t) dt \quad (\text{odd} \times \text{odd} = \text{even})$$

10.9 Transfer Functions and Linear Systems

Definition 58 (Transfer Function). *For a linear circuit, the transfer function $H(\omega)$ is defined as the ratio of the complex output to the complex input for a single sinusoidal component:*

$$H(\omega) \equiv \frac{\tilde{V}_{out}}{\tilde{V}_{in}}.$$

For a sinusoidal input $\tilde{V}_{in}(t) = V_0 e^{j\omega t}$, the circuit produces

$$\tilde{V}_{out}(t) = H(\omega) V_0 e^{j\omega t}.$$

For the RC low-pass circuit, $H(\omega) = \frac{Z_C}{Z_R + Z_C} = \frac{1}{1 + j\omega RC}$, giving

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}, \quad \phi(\omega) = -\arctan(\omega\tau).$$

For a real cosine input $V_{\text{in}}(t) = V_0 \cos(\omega t + \phi_0)$, the output is

$$V_{\text{out}}(t) = |H(\omega)| V_0 \cos(\omega t + \phi_0 + \phi(\omega)).$$

10.9.1 Application to a Sawtooth Input

Since the circuit is linear, it acts independently on each Fourier component. For a sawtooth input

$$V_{\text{in}}(t) = \sum_{n=1}^{\infty} b_n \sin(\omega_n t),$$

the output is

$$V_{\text{out}}(t) = \sum_{n=1}^{\infty} b_n |H(\omega_n)| \sin(\omega_n t + \phi(\omega_n)),$$

where

$$b_n = -\frac{(-1)^n}{n\pi}, \quad |H(\omega_n)| = \frac{1}{\sqrt{1 + (\omega_n\tau)^2}}, \quad \phi(\omega_n) = -\arctan(\omega_n\tau).$$

10.10 Complex Form of the Fourier Series

Using Euler's relation, the Fourier series can be rewritten in terms of complex exponentials:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)] = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_k t}, \quad \omega_k = \frac{2\pi k}{T},$$

where

$$c_0 = \frac{a_0}{2}, \quad c_m = \frac{a_m - jb_m}{2}, \quad c_{-m} = \frac{a_m + jb_m}{2}, \quad m \in \mathbb{N}.$$

The complex exponentials satisfy an orthogonality relation,

$$\int_{-T/2}^{T/2} e^{j\omega_k t} e^{-j\omega_{k'} t} dt = T \delta_{kk'},$$

from which the coefficients are extracted by

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\omega_k t} dt, \quad k \in \mathbb{Z}.$$

Example. Let $f(t) = \begin{cases} 1, & |t| \leq T/4 \\ 0, & T/4 < |t| \leq T/2 \end{cases}$. Then

$$c_k = \frac{1}{T} \int_{-T/4}^{T/4} e^{-j\omega_k t} dt = \frac{1}{2j\pi k} \left(e^{j\pi k/2} - e^{-j\pi k/2} \right) = \frac{\sin(\pi k/2)}{\pi k}.$$

The coefficients are real and symmetric ($c_k = c_{-k}$), which reflects the fact that $f(t)$ is a real-valued even function.

10.11 Fourier Series vs. Fourier Transform

	Fourier Series	Fourier Transform
Domain	Periodic $f(t)$ on finite interval	Aperiodic $f(t)$ on $(-\infty, \infty)$
Frequencies	Discrete: $\omega_k = \frac{2\pi k}{T}$	Continuous: $\omega \in \mathbb{R}$
Representation	$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_k t}$	$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$
Coefficients	$c_k = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\omega_k t} dt$	$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$

10.12 From Fourier Series to Fourier Transform

10.12.1 Negative Frequencies

In a Fourier series or Fourier transform, negative frequencies represent the clockwise rotation of complex phasors ($e^{-j\omega t}$), indicating a decreasing phase angle. They are the necessary mathematical counterpart to positive frequencies, ensuring that imaginary parts cancel when reconstructing real-valued signals.

10.12.2 The Limiting Argument

As the period T of a periodic function is stretched to infinity, the discrete harmonics in the frequency domain become more and more closely spaced, eventually forming a continuous function of frequency. This is the origin of the Fourier transform.

Define $\omega_k = \frac{2\pi k}{T}$, $c(\omega_k) \equiv c_k$, $F(\omega) \equiv Tc(\omega)$, and $\Delta\omega \equiv \frac{2\pi}{T}$. Then

$$F(\omega_k) = Tc(\omega_k) = \int_{-T/2}^{T/2} f(t) e^{-j\omega_k t} dt,$$

and the Fourier series can be rewritten as

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_k t} = \sum_{k=-\infty}^{\infty} \frac{F(\omega_k)}{T} e^{j\omega_k t} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} F(\omega_k) e^{j\omega_k t} \Delta\omega.$$

Taking the limit $T \rightarrow \infty$, the discrete spacing $\Delta\omega \rightarrow d\omega$ and the sum becomes an integral:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega.$$

A Fourier series uses a discrete sum because periodic functions contain only specific harmonics; the continuous integral of the Fourier transform acts as an orthogonal projection that isolates the contribution of each frequency.

10.13 Fourier Transform Pairs

The forward and inverse transforms form a symmetric pair:

$$f(t) \xrightarrow{\mathcal{F}} F(\omega) \quad (\text{forward}), \quad F(\omega) \xrightarrow{\mathcal{F}^{-1}} f(t) \quad (\text{inverse}).$$

In this nomenclature, the Fourier series coefficients c_k play the role of the inverse transform evaluated at discrete frequencies.

10.13.1 Example: Fourier Transform of a Box Function

Let $f(t) = 1$ for $|t| \leq a$ and zero otherwise. Then

$$F(\omega) = \int_{-a}^a e^{-j\omega t} dt = \frac{e^{j\omega a} - e^{-j\omega a}}{j\omega} = 2a \frac{\sin(\omega a)}{\omega a} = 2a \operatorname{sinc}(\omega a),$$

where $\operatorname{sinc}(x) \equiv \sin(x)/x$. The width of the box and the width of its Fourier transform are inversely related: a wider box produces a narrower sinc function, and a narrower box produces a wider sinc function.

10.14 Time-Bandwidth Product

The first zeros of $F(\omega) = 2a \operatorname{sinc}(\omega a)$ occur at $\omega = \pm\pi/a$. The width of the central peak in frequency space is therefore

$$\Delta\omega = \frac{\pi}{a} - \left(-\frac{\pi}{a}\right) = \frac{2\pi}{a}.$$

With $\Delta t = 2a$ denoting the width of the box in the time domain,

$$\Delta\omega \cdot \Delta t = 4\pi = \text{constant}.$$

This inverse relationship between time-domain and frequency-domain widths is a fundamental property of the Fourier transform, and is directly analogous to the Heisenberg uncertainty principle in quantum mechanics.

10.15 Dirac Delta Function

As the box width $a \rightarrow \infty$, the function $f(t) \rightarrow 1$ everywhere, and its Fourier transform $F(\omega) = 2a \operatorname{sinc}(\omega a)$ concentrates into a single spike at $\omega = 0$. The normalisation follows from

$$\int_{-\infty}^{\infty} F(\omega) d\omega = 2a \int_{-\infty}^{\infty} \frac{\sin(\omega a)}{\omega a} d\omega = 2\pi.$$

This motivates the definition of the **Dirac delta function**:

$$\delta(\omega) = \frac{1}{2\pi} \lim_{a \rightarrow \infty} 2a \operatorname{sinc}(\omega a),$$

with the properties

$$\delta(\omega) = \begin{cases} \infty, & \omega = 0 \\ 0, & \omega \neq 0 \end{cases}, \quad \int_{-\infty}^{\infty} \delta(\omega) d\omega = 1, \quad \int_{-\infty}^{\infty} \delta(\omega) G(\omega) d\omega = G(0).$$

10.16 Application: Pulse Input into an RC Circuit

Consider a rectangular voltage pulse of width Δt as the input to the RC low-pass circuit. Its Fourier transform is

$$\tilde{V}(\omega) = \int_{-\infty}^{\infty} V(t) e^{-j\omega t} dt = \Delta t \frac{\sin(\omega \Delta t / 2)}{\omega \Delta t / 2}.$$

Expressing $V(t)$ as a superposition of frequency components and applying the transfer function $H(\omega)$ to each component independently (by linearity),

$$V'(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) \tilde{V}(\omega) e^{j\omega t} d\omega = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega \Delta t / 2)}{\omega} \cdot \frac{e^{j\omega t}}{1 + j\omega\tau} d\omega,$$

which can be evaluated numerically. As the pulse width increases, the ringing in the Fourier transform becomes sharper, the amplitude of $\tilde{V}(\omega)$ grows, and the sinc function narrows.

10.17 Translated Dirac Delta Function

The Dirac delta function shifted to $\omega = \omega_0$ is defined by

$$\delta(\omega - \omega_0) = \begin{cases} \infty, & \omega = \omega_0 \\ 0, & \omega \neq \omega_0 \end{cases}, \quad \int_{-\infty}^{\infty} \delta(\omega - \omega_0) d\omega = 1, \quad \int_{-\infty}^{\infty} \delta(\omega - \omega_0) G(\omega) d\omega = G(\omega_0).$$

Its inverse Fourier transform is

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t},$$

i.e. a shifted delta function in frequency space corresponds to a pure complex exponential in time.

10.18 Fourier Transforms of Cosine and Sine

Using the translated delta function, the Fourier transforms of the elementary sinusoids follow directly.

Cosine. Writing $\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$ and comparing with the inverse transform,

$$f(t) = \cos(\omega_0 t) \implies F(\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)].$$

Sine. Writing $\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$,

$$f(t) = \sin(\omega_0 t) \implies F(\omega) = \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

Remark 16. Even though $\sin(x)$ is an odd function, the magnitudes of its positive and negative frequency components are still symmetric, because it is a real-valued function. Furthermore, $\sin(x)$ and $\cos(x)$ are the same function shifted by $\pi/2$; observe how this phase shift is encoded in the Fourier transform (the factor of $1/j$ corresponds to a $-\pi/2$ phase). The power spectrum $|F(\omega)|^2$ would lose this phase information.

10.19 Fourier Transform Pairs

The following table summarises the key Fourier transform pairs. The correspondence is symmetric: if $f(t) \leftrightarrow F(\omega)$, then $F(t) \leftrightarrow 2\pi f(-\omega)$, so each row applies in both directions.

Time domain $f(t)$	Frequency domain $F(\omega)$
Box function	sinc function (ringing)
Constant	Dirac delta $\delta(\omega)$
Complex exponential $e^{j\omega_0 t}$	Shifted delta $2\pi\delta(\omega - \omega_0)$
Cosine $\cos(\omega_0 t)$	Pair of deltas at $\pm\omega_0$ (real)
Sine $\sin(\omega_0 t)$	Pair of deltas at $\pm\omega_0$ (imaginary)
Dirac delta $\delta(t)$	Constant
sinc function	Box function
sinc^2 function	Triangle function
Gaussian	Gaussian

A constant function is a sinusoid with infinite period (zero frequency), which is why its transform is a single spike at $\omega = 0$. A cosine gives a purely real Fourier transform; a sine gives a purely imaginary one. In general, the Fourier transform of a real-valued function is complex.

10.20 Translation Theorems

Time translation. Shifting $f(t)$ by t_0 in the time domain multiplies its Fourier transform by a phase factor:

$$f(t - t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0} F(\omega).$$

The amplitude spectrum $|F(\omega)|$ is unchanged; only the phase is affected.

Frequency translation. Shifting $F(\omega)$ by ω_0 in the frequency domain corresponds to modulation in the time domain:

$$F(\omega - \omega_0) \xrightarrow{\mathcal{F}^{-1}} e^{j\omega_0 t} f(t).$$

10.21 Example: Fourier Transform of a Pulse Pair

Consider the sum of two identical square pulses of width Δt , displaced symmetrically about the origin:

$$f(t) = f_p(t - t_0) + f_p(t + t_0),$$

where f_p denotes a single square pulse with Fourier transform $F_p(\omega)$. By the time-translation theorem,

$$F(\omega) = e^{-j\omega t_0} F_p(\omega) + e^{j\omega t_0} F_p(\omega) = 2F_p(\omega) \cos(\omega t_0).$$

The result is a sinc function modulated by a cosine, producing an interference (beat) pattern in the frequency domain. This is directly analogous to the fringe pattern produced by two radio telescopes operating as an interferometer: the two spatially separated apertures play the role of the two time-displaced pulses.

10.22 Reality Condition

The Fourier transform can in general be complex, but the requirement that $f(t)$ be real-valued imposes a constraint on $F(\omega)$. If $f(t) \in \mathbb{R}$, then $f(t) = f^*(t)$, and it follows that

$$F^*(\omega) = F(-\omega).$$

That is, the Fourier transform of a real signal has Hermitian symmetry: the negative-frequency content is entirely determined by the positive-frequency content.

10.23 Power Spectrum

Definition 59 (Power Spectrum). *The power spectrum of a signal $f(t)$ is*

$$|F(\omega)|^2 = F(\omega) F^*(\omega) = F(\omega) F(-\omega).$$

The square (rather than the square root) is used because power is proportional to the square of amplitude, $P \propto A^2$.

When plotting $|F(\omega)|^2$, phase information is lost and the spectrum is symmetric in ω (since $|F(-\omega)|^2 = |F(\omega)|^2$ for real signals).

10.23.1 One-Sided Spectrum

Since the power spectrum is symmetric, it is common to plot the **one-sided spectrum**: $2|F(\omega)|^2$ over positive ω only. The factor of two ensures that the total integrated power remains equal to that of the original two-sided spectrum, since the power from the discarded negative frequencies is folded back onto the positive side.

Example. For $f(t) = \sin(\omega_0 t)$, the Fourier transform is

$$F(\omega) = \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)],$$

giving a power spectrum with two symmetric spikes at $\pm\omega_0$.

10.24 Finite-Length Signals and Windowing

The Fourier transform of a finite-duration signal is not an exact Dirac delta function: a finite sine wave has a defined duration in time, and therefore cannot be perfectly localised to a single frequency. In practice, a finite-length signal $f(t)$ is represented as the product of an infinite-duration signal $s(t)$ and a **window function** $w(t)$:

$$F(\omega) = \int_{-\infty}^{\infty} w(t) s(t) e^{-j\omega t} dt.$$

Multiplying the signal by a smooth window function suppresses the side lobes that arise from the sharp truncation, at the expense of a modest increase in bandwidth (i.e. a broadening of the central peak).

10.25 Discrete Sampling and the Nyquist Criterion

Experimental signals are necessarily measured at discrete time points $t_n = n \Delta t_s$, where n is an integer and Δt_s is the sampling interval. The corresponding **sampling frequency** is $f_s = 1/\Delta t_s$.

Theorem 7 (Nyquist Criterion). *To faithfully reconstruct a signal, the sampling frequency must be at least twice the maximum frequency present in the signal:*

$$f_s \geq 2f_{\max} \implies f_{\max} \leq \frac{f_s}{2} = \frac{1}{2\Delta t_s}.$$

The maximum detectable frequency $f_{\max} = f_s/2$ is called the **Nyquist frequency**.

Example. To detect a pulsar with period $P = 1$ ms, the time resolution must be at most $\Delta t_s = 0.5$ ms. To also resolve its fourth harmonic (period $P/4 = 0.25$ ms), the sampling interval must be reduced to $\Delta t_s \leq 0.125$ ms.

10.26 Discrete Fourier Transform

For a discretely sampled signal of N points, the **Discrete Fourier Transform (DFT)** is

$$F(k) = \frac{1}{N} \sum_{n=0}^{N-1} f(n) e^{-j2\pi nk/N}.$$

For a discrete sine wave with sampling interval Δt , period P , and total observation time $T_{\max} = N \Delta t$, the DFT produces:

- A peak at the fundamental angular frequency $\omega_0 = 2\pi/P$.
- A frequency resolution of $\Delta\omega \approx 2\pi/T_{\max}$ (finer resolution requires longer observation).
- A maximum detectable frequency of $\omega_{\max} \approx \pi/\Delta t$ (the Nyquist frequency in angular units).

10.27 Amplitude Modulation

Definition 60 (Amplitude Modulation (AM)). *Amplitude modulation encodes a signal by varying the amplitude of a high-frequency carrier wave. If the carrier is $\cos(\omega_c t)$ and the baseband signal is $f(t)$, the modulated signal is*

$$s(t) = [1 + f(t)] \cos(\omega_c t).$$

By the frequency-translation theorem, modulation shifts the spectrum of $f(t)$ to be centred on $\pm\omega_c$. AM radio uses this principle to transmit audio signals over radio-frequency carriers.

11 Appendix

12 Useful Links

Inductors (visual explanation): <https://www.youtube.com/shorts/lXk4CQDbr4Q>
Capacitors (visual explanation): <https://www.youtube.com/shorts/PYrTLQctdwg>
Resistors (introduction): <https://www.youtube.com/watch?v=5HI33hSZoH4>
Drawing Circuits for latex: <https://www.circuit2tikz.tf.fau.de/designer/>